

Chap 14 14.3 Partial Derivatives

Partial Derivative with Respect to x

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

Mixed Derivative Theorem :

if $f(x, y)$ & f_x, f_y, f_{xy}, f_{yx} are all defined throughout an open region containing (a, b) & are all continuous at (a, b) then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

14.4 The Chain Rule

Chain Rule for Two independent variables

$$\frac{df}{dt} = f_x(x(t), y(t)) \cdot x'(t) + f_y(x(t), y(t)) \cdot y'(t)$$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Chain Rule for Three Independent Variables

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}$$

Chain Rule - Two independent variables &
Three Intermediate Variables

$$w = f(x, y, z), \quad x = g(r, s), \quad y = h(r, s), \quad z = k(r, s)$$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial r}$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s}$$

Implicit Differentiation:

$$\frac{dy}{dx} = - \frac{F_x}{F_y}$$

14.5 Directional Derivatives & Gradient Vectors

Directional Derivative - in the direction

$$\left(\frac{df}{ds} \right)_{u, p_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

Gradient Vector: $f(x, y)$ at point $p_0(x_0, y_0)$

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

Directional Derivative:

$$\left(\frac{df}{ds}\right)_{u,p_0} = (\nabla f)_{p_0} \cdot u$$

Properties of directional derivative

$$D_u f = \nabla f \cdot u = |\nabla f| \cos \theta$$

1. $\uparrow$ most rapidly when $\cos \theta = 1$ / when u is the direction of ∇f . $\cos(0)$

2. $\downarrow$ most rapidly in the direction of $-\nabla f$.
 $D_u f = |\nabla f| \cos(\pi) = -|\nabla f|$

3. u orthogonal to a gradient $\nabla f \neq 0$

$$D_u f = |\nabla f| \cos(\pi/2) = |\nabla f| \cdot 0 = 0$$

14.6 Tangent Planes & Differentials

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$x = x_0 + f_x(P_0)t, \quad y = y_0 + f_y(P_0)t, \quad z = z_0 + f_z(P_0)t$$

Plane tangent to a surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$

Estimating change in f in a direction u :

$$df = (\nabla f|_{P_0} \cdot u) \cdot ds \leftarrow \text{distance increment}$$

 Directional derivative

Linearization, Standard linear approximation

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$f(x, y) \approx L(x, y)$$

Error in Standard Linear Approx.

rectangle R centered at (x_0, y_0)

M is upper bound for values of $|f_{xx}|, |f_{yy}|, |f_{xy}|$

$$|E(x, y)| \leq \frac{1}{2} M (|x - x_0| + |y - y_0|)^2$$

Total Differential:

$$df = f_x(x_0, y_0) dx + f_y(x_0, y_0) dy$$

14.7 Extreme Values and Saddle Points

First derivative test for local extreme values

$$f_x(a,b) = 0 \quad \& \quad f_y(a,b) = 0$$

Critical Point:

$f_x \& f_y = 0$ / one of $f_x \& f_y$ or both don't exist
 $\Rightarrow$ critical point of f

Saddle point:

Second derivative test for local extreme values

- i) local max $\Rightarrow f_{xx} < 0$, $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b)
- ii) local min $\Rightarrow f_{xx} > 0$, $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b)
- iii) saddle point $\Rightarrow f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a,b)
- iv) inconclusive $\Rightarrow f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a,b)

$$f_{xx}f_{yy} - f_{xy}^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$$